Recursive self-improvement of AI research agents

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HarnessRSI 64 ★★

evidence: retained

Paper

1 What improves what?

Start: research-agent harness
AIDE0: greedy tree search + full history
Fixed model weights; mutable agent code

▼

Outer agent proposes a rewrite
AIDEhuman / Claude Opus 4.7
Edits incumbent; reads past grades

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Candidate does research tasks
Gemini 3 Flash; fixed dollars per task
Public feedback guides solution search

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Private outcomes give grade g
Score the solutions the agent returns
Average over tasks and repeated runs

▼

Keep only the best-graded agent
Accepted code is the next edit target
Buggy proposals get no grade

Retained in AIDE85:
bandit + forks · bounded context
error memory · robustness changes

The outer loop sees private grades; only the inner task agent is blind to private data. Selection g is not an untouched test.

Finite trajectory + task budgets stop the run. No statistical acceptance gate or runaway-safety guarantee.

2 Transfer, not four decisive wins

ALE · contest performance

AIDE0	1536 ± 33
AIDE85	1790 ± 9
human	1511 ± 35

MLE · private percentile

AIDE0	0.678 ± 0.006
AIDE85	0.722 ± 0.011
human	0.708 ± 0.007

Weather · skill gain

AIDE0	0.262 ± 0.205
AIDE85	0.793 ± 0.005
human	0.404 ± 0.193

FML · test improvement (%)

AIDE0	15.0 ± 0.9
AIDE85	19.9 ± 1.1
human	19.6 ± 1.0

Table 1: means ±1 SE, separate zero-based scales. Checkpoints from the featured RSI lineage, not means across RSI runs.

ALE: 10 tasks ×10 seeds; MLE: 22 ×3; Weather: 1 ×3; FML: 18 ×3. Weather is one optimization task, not a leaderboard score.

4 Where gains thin out

- **FML & MLE:** gaps to human are small relative to SEs (chart above); superiority not established.
- **Not monotone:** AIDE47 beats AIDE85 on MLE (0.730 vs 0.722) and Weather (0.798 vs 0.793).
- **Model transfer:** $20 ALE/MLE tests use Gemini, Fable, GPT. Fable MLE improvement is within one SE (App. C).
- No per-task breakdown or component ablation. Robust selection penalty never changes ranking in replay.

3 Controls & the ignition test

Repeated improvement ≠ acceleration

8 days · 100 nodes · 99 proposals; seven kept, g: 0.703 → 0.778. Other full runs keep two / four; featured lineage tested externally.

Same evaluation protocol

- All agents: ALE/MLE $5 (Gemini 3 Flash); Weather $15 (Gemini 3.1 Pro); FML 100 steps (GPT-5.4), not equal dollars.
- Tasks excluded from selection; Weather family unseen. No pretraining contamination audit.
- Gemini and defaults selected on selection tasks; separate tuning split / baseline tuning budget unreported.

Can it improve the improver?

Outer agent	Endpoint mean
AIDE47	0.780
AIDEhuman	0.782

Same AIDE47 start; 3 RSI seeds/arm, 50 steps (Fig. 7 caption). Min–max, not CIs. Inconclusive.

Behavior & missing controls

- Kernel hacking: 55% → 32%; human 39%. 38 pairs, 3-seed speedup means; no rate CI.
- Scoring-script repair: no same-repair baseline ablation. Execution metrics judge research; LLM extracts feedback.
- No total RSI dollar/token bill; evaluation caps exclude discovery cost.

Takeaway

Search can discover a research harness that transfers beyond its selection tasks. But a better researcher is not yet a demonstrably better self-improver: the ignition test is inconclusive, external gains are non-monotone, and discovery cost remains partly quantified.

Our read · HarnessRSI

64 · ★★ · evidence: retained

Reports: external tests, seeds + SEs, matched protocols, model transfer.

Not established: better improver, component causality, acceleration. Variance judgment understated; see SOURCE.